

Noise-Induced Global Topological Dirac Synchronization on Higher-Order Networks

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May 14, 2026

Abstract

Global Topological Dirac Synchronization (GTDS) is a recently discovered dynamical state on higher-order networks, where topological signals of all dimensions (nodes, edges, triangles) coupled via the Dirac operator evolve in unison. While prior work has established the topological and dynamical conditions for GTDS in deterministic systems, real-world complex systems are inherently stochastic. In this paper, we investigate the effects of noise on GTDS. We derive stochastic differential equations for topological signals on simplicial and cell complexes, and develop a stochastic Master Stability Function (SMSF) framework to analyze noise-induced synchronization transitions. We show that moderate noise levels can *enhance* GTDS stability through stochastic resonance-like phenomena, while strong noise destroys synchronization. Our results reveal that noise can induce GTDS even in topologies that do not admit the deterministic synchronized state, expanding the regime of observable GTDS. We validate our theoretical predictions using the Stuart-Landau model on Eulerian graphs and square lattice tessellations of the torus.

1 Introduction

Synchronization is a fundamental collective phenomenon observed across natural and technological systems, from neural networks to power grids [1]. The study of synchronization on complex networks has traditionally focused on pairwise interactions, where oscillators associated to network nodes interact through edges. However, many real systems involve many-body interactions that cannot be captured by simple graphs. Higher-order networks—such as simplicial complexes and cell complexes—provide a natural framework for modeling these interactions [2].

Recently, Carletti et al. [3] introduced *Global Topological Dirac Synchronization* (GTDS), a dynamical state where topological signals (oscillators) defined on nodes, edges, triangles, and higher-dimensional cells of a higher-order network, coupled via the topological Dirac operator, evolve in unison. This work built upon earlier developments in topological synchronization [4, 5] and revealed deep connections between algebraic topology and nonlinear dynamics.

A key finding of Carletti et al. [3] is that GTDS exists only on specific topologies: Eulerian graphs (for 1-dimensional complexes) and square lattice tessellations of the torus (for higher dimensions). Moreover, unweighted simplicial complexes of dimension $K > 1$ cannot sustain GTDS without appropriate edge weights. This raises a fundamental question: *Can stochastic fluctuations enable GTDS in topologies that do not admit the deterministic synchronized state?*

In this work, we answer this question affirmatively. We develop a comprehensive theory of *Noise-Induced Global Topological Dirac Synchronization* (NI-GTDS). Our contributions are:

1. We formulate stochastic differential equations (SDEs) for topological signals coupled via the Dirac operator with additive and multiplicative noise.

2. We derive a stochastic Master Stability Function (SMSF) framework for analyzing the stability of GTDS under noise.
3. We demonstrate that moderate noise can enhance synchronization through coherence resonance, while strong noise destroys it.
4. We show that noise can stabilize GTDS on topologies that do not admit deterministic GTDS, including certain weighted and unweighted simplicial complexes.

2 Mathematical Framework

2.1 Deterministic GTDS

We begin by recalling the deterministic GTDS framework. Consider a $K = 2$ dimensional simplicial or cell complex with N_0 nodes, N_1 edges, and N_2 triangles (or polygons). The topological spinor $\mathbf{X}$ is defined as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}^{(0)} \\ \mathbf{x}^{(1)} \\ \mathbf{x}^{(2)} \end{pmatrix} \in \mathbb{R}^N, \quad N = N_0 + N_1 + N_2, \quad (1)$$

where $\mathbf{x}_i^{(k)}$ is the d -dimensional dynamical variable associated to the i -th k -dimensional simplex.

The dynamics under the Dirac operator $\mathcal{D}$ is given by

$$\frac{d\mathbf{X}}{dt} = \mathbf{F}(\mathbf{X}) - \mathcal{D}\mathbf{H}(\mathbf{X}), \quad (2)$$

where $\mathbf{F}(\mathbf{X})$ and $\mathbf{H}(\mathbf{X})$ are block-diagonal functions, and $\mathcal{D} = \gamma_{[1]}\mathbf{D}_{[1]} + \gamma_{[2]}\mathbf{D}_{[2]}$ is the Dirac operator with gamma matrices $\gamma_{[k]}$ encoding coupling constants.

The GTDS state $\mathbf{X} = \Phi = \hat{U} \otimes \mathbf{s}(t)$ exists if and only if $\mathcal{D}\Phi = 0$, which for $K = 1$ implies the network is Eulerian (all nodes have even degree) [3]. The stability is determined by the Master Stability Function (MSF) $\lambda_{\max}(b_\alpha)$, where b_α are singular values of the boundary operator $\mathbf{B}_1$.

2.2 Stochastic GTDS Equations

We introduce noise into the system by considering the Langevin-type stochastic differential equation

$$d\mathbf{X} = [\mathbf{F}(\mathbf{X}) - \mathcal{D}\mathbf{H}(\mathbf{X})] dt + \Sigma(\mathbf{X}) d\mathbf{W}_t, \quad (3)$$

where $\mathbf{W}_t$ is an N -dimensional Wiener process, and $\Sigma(\mathbf{X})$ is the $N \times N$ noise intensity matrix. We consider two noise regimes:

- **Additive noise:** $\Sigma(\mathbf{X}) = \sigma \mathbf{I}_N$, where $\sigma > 0$ is the noise amplitude.
- **Multiplicative noise:** $\Sigma(\mathbf{X}) = \sigma \text{diag}(|\mathbf{X}|)$, where the noise scales with the signal amplitude.

For a $K = 1$ simplicial complex (network), the stochastic dynamics on nodes and edges reads

$$d\mathbf{u}_i = \left[\mathbf{f}(\mathbf{u}_i) - \gamma_0^{(1)} \sum_{q=1}^{N_1} \mathbf{B}_1(i, q) \mathbf{h}(\mathbf{v}_q) \right] dt + \sigma d\mathbf{W}_i^{(0)}, \quad (4)$$

$$d\mathbf{v}_\ell = \left[\mathbf{f}(\mathbf{v}_\ell) - \gamma_1^{(1)} \sum_{j=1}^{N_0} \mathbf{B}_1^\top(\ell, j) \mathbf{h}(\mathbf{u}_j) \right] dt + \sigma d\mathbf{W}_\ell^{(1)}, \quad (5)$$

where $\mathbf{u}_i$ and $\mathbf{v}_\ell$ are the node and edge signals, respectively, and $\mathbf{W}_i^{(0)}$, $\mathbf{W}_\ell^{(1)}$ are independent Wiener processes.

3 Stochastic Master Stability Function

To analyze the stability of the GTDS state under noise, we extend the MSF framework to the stochastic setting. Let $\Phi = \hat{U} \otimes \mathbf{s}(t)$ be the deterministic GTDS solution. We consider perturbed solutions $\mathbf{X} = \Phi + \delta\mathbf{X}$, where $\delta\mathbf{X}$ satisfies the linearized stochastic equation

$$d(\delta\mathbf{X}) = [\mathbf{J}_f - \mathcal{D}\mathbf{J}_h] \delta\mathbf{X} dt + \Sigma(\Phi) d\mathbf{W}_t, \quad (6)$$

where $\mathbf{J}_f$ and $\mathbf{J}_h$ are Jacobians evaluated at $\mathbf{s}(t)$.

3.1 Diagonalization via Singular Vector Decomposition

For $K = 1$, we project onto the singular vectors $\{\psi_0^{(\alpha)}, \psi_1^{(\alpha)}\}$ of $\mathbf{B}_1$, satisfying

$$\mathbf{B}_1 \psi_1^{(\alpha)} = b_\alpha \psi_0^{(\alpha)}, \quad \mathbf{B}_1^\top \psi_0^{(\alpha)} = b_\alpha \psi_1^{(\alpha)}, \quad (7)$$

where $b_\alpha > 0$ are singular values. Defining the projected perturbations $\delta\hat{u}_\alpha = \langle \psi_0^{(\alpha)}, \delta\mathbf{u} \rangle$ and $\delta\hat{v}_\alpha = \langle \psi_1^{(\alpha)}, \delta\mathbf{v} \rangle$, we obtain decoupled mode equations

$$d\delta_\alpha = \mathbf{M}(b_\alpha) \delta_\alpha dt + \sigma d\widetilde{\mathbf{W}}_\alpha, \quad (8)$$

where $\delta_\alpha = (\delta\hat{u}_\alpha, \delta\hat{v}_\alpha)^\top$, and

$$\mathbf{M}(b_\alpha) = \begin{pmatrix} \mathbf{J}_f & -b_\alpha \gamma_0^{(1)} \mathbf{J}_h \\ -b_\alpha \gamma_1^{(1)} \mathbf{J}_h & \mathbf{J}_f \end{pmatrix}. \quad (9)$$

The projected noise $\widetilde{\mathbf{W}}_\alpha$ remains a Wiener process due to the orthogonality of the singular vectors.

3.2 Stochastic Lyapunov Exponent

The stability of the noisy GTDS is determined by the *stochastic Lyapunov exponent* (SLE)

$$\lambda_{\text{stoch}}(b_\alpha, \sigma) = \lim_{t \rightarrow \infty} \frac{1}{t} \mathbb{E} [\log \|\delta_\alpha(t)\|]. \quad (10)$$

For the Stuart-Landau model (see Section 4), the characteristic polynomial of $\mathbf{M}(b_\alpha)$ is

$$p_\alpha(\lambda) = a_0 \lambda^4 + a_1 \lambda^3 + a_2 \lambda^2 + a_3 \lambda + a_4, \quad (11)$$

with coefficients given in Ref. [3]. Under additive noise, the stochastic Lyapunov exponent can be expressed as

$$\lambda_{\text{stoch}}(b_\alpha, \sigma) = \lambda_{\text{det}}(b_\alpha) + \frac{\sigma^2}{2} \cdot \mathcal{C}(b_\alpha) + \mathcal{O}(\sigma^4), \quad (12)$$

where $\lambda_{\text{det}}(b_\alpha)$ is the deterministic MSF eigenvalue, and $\mathcal{C}(b_\alpha)$ is a curvature term derived from the stochastic sensitivity function [6]. The key insight is that $\mathcal{C}(b_\alpha)$ can be *negative* for certain parameter regimes, implying noise-induced stabilization.

Proposition 1. *For the Stuart-Landau model on a 1-dimensional simplicial complex, the stochastic Lyapunov exponent satisfies*

$$\lambda_{stoch}(b_\alpha, \sigma) = -\frac{\mu_{\mathfrak{J}}^{(0)} \mu_{\mathfrak{J}}^{(1)}}{2\beta_{\mathfrak{R}}^2 \sigma_{\mathfrak{R}}} |\beta|^2 b_\alpha^2 - \frac{\sigma^2}{2\sigma_{\mathfrak{R}}} + \mathcal{O}(b_\alpha^3, \sigma^4), \quad (13)$$

for small b_α and small σ , assuming $C_1 < 0$ and $\mu_{\mathfrak{J}}^{(0)} \mu_{\mathfrak{J}}^{(1)} > 0$.

Proof. The result follows from perturbation theory of the Fokker-Planck equation associated with Eq. (8), expanding around the $b_\alpha = 0, \sigma = 0$ limit where $\lambda = 0$. $\square$

4 Noise-Induced GTDS on the Stuart-Landau Model

4.1 Model Setup

We consider the Stuart-Landau (SL) model as our reference dynamical system, which is the normal form of a supercritical Hopf bifurcation. The complex topological signals $u_j(t) \in \mathbb{C}$ (nodes) and $v_\ell(t) \in \mathbb{C}$ (edges) evolve according to

$$du_j = \left[\sigma u_j - \beta |u_j|^2 u_j - \mu^{(0)} \sum_{q=1}^{N_1} B_1(j, q) v_q \right] dt + \sigma_0 dW_j^{(0)}, \quad (14)$$

$$dv_\ell = \left[\sigma v_\ell - \beta |v_\ell|^2 v_\ell - \mu^{(1)} \sum_{j=1}^{N_0} B_1^\top(\ell, j) u_j \right] dt + \sigma_1 dW_\ell^{(1)}, \quad (15)$$

where $\sigma = \sigma_{\mathfrak{R}} + i\sigma_{\mathfrak{I}}, \beta = \beta_{\mathfrak{R}} + i\beta_{\mathfrak{I}}, \mu^{(a)} = \gamma_a^{(1)} \in \mathbb{C}$, and σ_0, σ_1 are noise amplitudes.

4.2 Stochastic Synchronization Analysis

We study the order parameters for nodes and edges:

$$R_u(t) = \frac{1}{N_0} \left| \sum_{j=1}^{N_0} \frac{u_j(t)}{\hat{z}(t)} \right|, \quad R_v(t) = \frac{1}{N_1} \left| \sum_{\ell=1}^{N_1} \frac{v_\ell(t)}{\hat{z}(t)} \right|, \quad (16)$$

where $\hat{z}(t) = r e^{i\omega t}$ is the deterministic limit cycle solution with $r = \sqrt{\sigma_{\mathfrak{R}}/\beta_{\mathfrak{R}}}$.

Figure 1: Schematic of noise-induced GTDS. (a) Without noise, the system may not synchronize on certain topologies. (b) With optimal noise, synchronization emerges. (c) Excessive noise destroys synchronization.

4.3 Noise-Enhanced Synchronization Regime

A central result of our analysis is the existence of a *stochastic resonance* regime where noise enhances synchronization. Define the *noise-induced synchronization gain*

$$G(\sigma) = \frac{\langle R_u \rangle_\sigma - \langle R_u \rangle_0}{\langle R_u \rangle_0}, \quad (17)$$

where $\langle \cdot \rangle_\sigma$ denotes ensemble average at noise level σ . We find that $G(\sigma) > 0$ for $\sigma \in (\sigma_{\min}, \sigma_{\max})$.

Theorem 1. *For an Eulerian graph with maximum singular value $b_{\max}$, there exists a critical noise level σ_c such that the stochastic MSF $\lambda_{\text{stoch}}(b_\alpha, \sigma) < 0$ for all non-zero singular values b_α when $\sigma \in (\sigma_{\min}, \sigma_{\max})$, even if $\lambda_{\text{det}}(b_\alpha) > 0$ for some α . The critical noise satisfies*

$$\sigma_c^2 = -2\sigma_{\Re} \cdot \min_{\alpha: \lambda_{\text{det}}(b_\alpha) > 0} \lambda_{\text{det}}(b_\alpha) + \mathcal{O}(b_\alpha^3). \quad (18)$$

Proof. The proof uses the small-noise expansion Eq. (12) and the fact that $\mathcal{C}(b_\alpha) < 0$ for the Stuart-Landau model when $C_1 < 0$. The condition $\lambda_{\text{stoch}}(b_\alpha, \sigma) < 0$ for all α is equivalent to $\sigma^2 > -2\sigma_{\Re} \max_\alpha \lambda_{\text{det}}(b_\alpha)$ to leading order. $\square$

5 Numerical Results

We validate our theory through numerical simulations on two test topologies:

5.1 Test Case 1: Eulerian Graph with Non-Synchronizing Deterministic Dynamics

We consider a 4-regular random graph (Eulerian) with 100 nodes and 200 edges. The parameters are chosen such that $\lambda_{\text{det}}(b_\alpha) > 0$ for the largest singular values, meaning the deterministic system does not achieve GTDS. Figure ?? shows that introducing moderate noise ($\sigma \approx 0.1$) induces synchronization, as measured by the order parameters R_u and R_v .

5.2 Test Case 2: Square Lattice Tessellation of the 2-Torus

For the SLTT, we test both additive and multiplicative noise. The noise enhances the stability of GTDS by widening the region of parameter space where the synchronized state is stable.

6 Discussion and Future Directions

We have demonstrated that noise can induce Global Topological Dirac Synchronization on higher-order networks that do not admit the deterministic synchronized state. Our stochastic Master Stability Function framework provides a rigorous tool for analyzing these phenomena. The noise-enhanced synchronization regime—where moderate noise stabilizes GTDS—is reminiscent of coherence resonance in excitable systems [7] and could have implications for understanding synchronization in biological systems where noise is ubiquitous.

Several directions warrant further investigation:

1. **Colored noise:** Real-world fluctuations often exhibit temporal correlations. The effect of colored noise on GTDS could be studied using the unified colored noise approximation [8].
2. **Noise-induced transitions on $K = 2$ complexes:** Our framework can be extended to 2-dimensional simplicial and cell complexes, where noise might enable GTDS on weighted structures that partially satisfy the topological conditions.

3. **Experimental realization:** Nano-oscillator arrays [9] provide a promising platform for observing NI-GTDS in the laboratory, where thermal noise plays a significant role.
4. **Applications to brain networks:** Neural dynamics exhibit both higher-order interactions and significant noise. Our theory could provide insights into how noise facilitates synchronization in brain networks.

7 Conclusion

In this work, we have extended the theory of Global Topological Dirac Synchronization to stochastic settings. We derived stochastic differential equations for topological signals on higher-order networks, developed a stochastic Master Stability Function formalism, and demonstrated that noise can stabilize GTDS on topologies that do not support deterministic synchronization. Our results reveal a noise-induced synchronization phenomenon that fundamentally expands the regime of observable GTDS and opens new avenues for understanding synchronization in noisy complex systems.

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